Assessment Tool - 3

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1.) PEAS Description and intelligent Agent

a.) Performance measure:-

-> cleanliness of both rooms (maximizing dirt Pickup
-> minimizing energy (power consumption
-> minimizing operation time.

environment:-

-> Two rooms (Room A and Room B) connected.
-> Each room can be in one of two states: clean or dirtly

Actuators:-

->wheels (motors enabling movement left or Right (to switch rooms)
-> Suction mechanism to suck dirt
-> A do nothing (no-op) action when no teake is required.

Sensors

-> Dirt Sensor to detect whether the current room is clean or dirty
-> Location sensor to identify which room (Left (Right) the robot is currently in

b.) identify the most suitable intelligent Agents

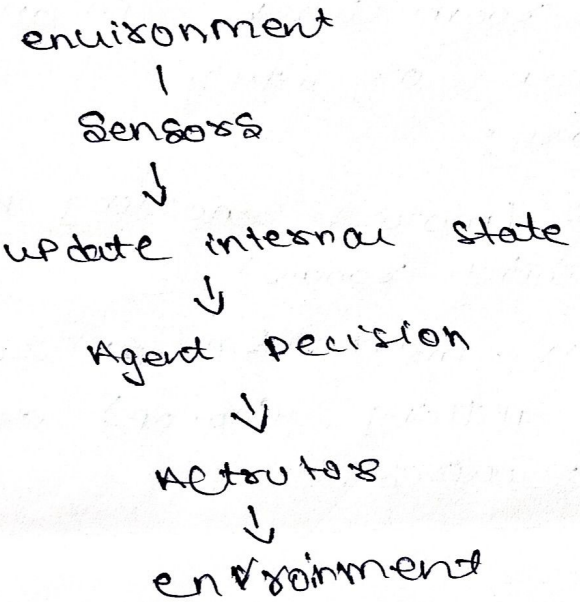
-> model - Based Reflex Agent

-> A model Based reflex agent is suitable for this scenario when the robot must remembers information about the two room.

Working :-

-> The robot maintains an internal state containing
  -> Its current location.
  -> The Cleanliness state of the first room.
  -> The Cleanliness state of the second room.

working diagram :-



Conclusion :-

A model - based reflex agent is suitable choice when the robot needs internal memory to track both rooms.

2 Task environments properties Table:

Property environment	playing	Self Driving Car	Crossword Puzzle
Observability	Fully observable	Partially observable	Fully observable
Determinism	Deterministic	Stochastic	Deterministic
Episodic vs Sequential	Sequential	Sequential	Sequential
Static vs Dynamic	Static	Dynamic	Static
Discrete vs Continuous	Discrete	Continuous	Discrete.

3) Formal Search problem formulation for a Delivery Robot

(i) Initial state :-

-> The Robot's starting location coordinate inside the building grid.

(ii) Goal State.

-> The destination delivery point the test checks if robot's current position matches the delivery cell.

(iii) Action :-

-> The set of allowed movements up Down Left Right.

(iv) Transition model:

-> Return the resulting state when a specific action is applied from a current grid cell provided the cell is not blocked by an obstacle.

(v) path cost fn:

-> the total cost of the path usually measured by the no of steps taken or distance traveled.

4.) Breadth First Search (BFS)

BFS : BFS visits nodes level by level using a queue (FIFO)

step	node visited	queue
1	S	A, B
2	A	B, C, D
3	B	C, D, E
4	C	D, E, F
5	D	E, F
6	E	F, G
7	F	G
8	G	empty

Final path : S - B - E - G

path cost : 3 edges

properties of BFS

-> Time complexity : O(V+E)
space complexity : O(V)

5] Depth - First Search (DFS) :-

DFS :- DFS explores one branch deeply before backtracking using a stack (LIFO)
Assuming the left branch is explored first.

Step	node visited	stack
1	S	B, A
2	A	B, D, C
3	C	B D f
4	F	B D
5	D	B
6	B	E
7	E	G
8	G	Empty

Final path : S - B - E, G

path : 3 edges

Comparison and conclusion :-

-> BFS finds the shortest path in an unweighted graph.
-> DFS may find a path depending on the traversal order.

Conclusion :-

BFS is more suitable when the shortest path is required because it guarantees the shortest path in an unweighted graph.